

AEL - Technical Blueprint

Group 3 - Tanishq Bhaiji, Nikola Hadzhiminov, Valentina Petrušić, Tanjin Rahman Rafi, Anisah Sakura Sutrisno

Implements the PRD (revision 2), Target SME: Thuiszorg Totaal, Arnhem

1. The parts

Eight parts, drawn as a production line (stages) with a shared context file every stage reads. Each part does one job, describable in one sentence.

Part	Job (one sentence)	What it produces
Context	Holds what every other part reads before it starts	A single context file: target SME (Thuiszorg Totaal, ~70 staff, Arnhem), six quality criteria from PRD §3, house style, what is out of scope
Intake	Accepts the week's raw material and normalises it	Files in /sources/raw/ with metadata (origin, language, date). Sources include interview notes, CBS/KvK data, press, sector material
Translate	Turns Dutch audio into Dutch transcript, then translates Dutch sources to English; keeps both side by side	Paired files: source.nl.md + source.en.md, linked. Original retained for verification (PRD §4)
Retrieve	Runs search queries in both Dutch and English for the week's research theme	Adds results to /sources/raw/ with the same metadata contract
Appraise	Writes a credibility note per source: publisher, recency, relevance, reliability, keep/discard	One appraisal.md per source in /sources/appraised/
Draft	Assembles one chapter in the handbook's fixed anatomy from appraised sources	draft.md in /drafts/, every claim tagged with its source file
Check	Holds the draft against the six quality criteria (clear, practical, easy to understand, evidence-based, responsible, actionable) before a human sees it	check-report.md: pass/fail per criterion, with the failing line quoted
Publish	Pushes the approved chapter to the existing handbook site	Copies final.md to the site's expected location and format

Decision 1: parts are stages on a path from source to published page, reading a shared context file. Decision 2: an orchestrator (the managing-editor prompt) calls each stage in order and pauses for human inspection after each. Decision 4: shared context lives in one file every part reads.

Who runs which part

The PRD assigns five people to four roles. Here is how each role maps to the pipeline:

Role	Person	Which parts they drive
Product Manager	Anisah	Writes the weekly research task brief (topic, priority, acceptance criteria). Signs off before publication. Does not run the CLI pipeline directly.
Developers	Tanishq, Nikola	Translate the PM's brief into CLI prompts that drive Intake, Translate, Retrieve, Appraise and Draft. Refine prompts when the Tester flags issues.
Tester	Valentina	Runs Check: holds every stage's output against its expected shape and the six quality criteria before the workflow continues.
Deployer	Tanjin (+ Nikola)	Runs Publish: gets the approved chapter onto the handbook site and verifies it is accessible and correctly formatted.

Every part is runnable by anyone on the team from a checklist. The role table says who is *accountable* for a part, not who is *capable* of it.

2. What passes between the parts

The contract between two parts is a file with a fixed shape. This gives us the manual fallback for free: any file a part produces, a person can produce by hand, and the next part cannot tell the difference. Three seams are named:

Seam	In	Out	Why this one
Intake → Translate	Raw Dutch audio or document in /sources/raw/	Paired .nl.md + .en.md in /sources/translated/	The Dutch gap seam. This is what makes language stop deciding who researches.
Appraise → Draft	Appraised source files in /sources/appraised/	draft.md in /drafts/, every claim tagged [source: filename]	The evidence seam. A claim without a source tag does not pass Check.
Check → human reviewer	check-report.md listing pass/fail per criterion	Approved final.md, or feedback for a re-draft	The human gate. Check is automated; approval is never automated (PRD §5).

Decision 3: the contract is a file with a fixed shape — named, with named fields.

3. The line nothing may cross

Interview recordings, transcripts and any material identifying Thuiszorg Totaal's clients by name do not leave the platform's internal folder and do not reach a model service. Only Translate and Retrieve may reach the open web and any part may call the LLM API, but client-identifying fields are stripped before any call leaves the orchestrator. Interview material enters the platform and is stopped at the one exit it may not use. Consent forms and anonymisation decisions are carried by a person, not by the platform.

Decision 5: one part (the orchestrator) controls traffic to the outside world, and the rule about what may not pass is enforced there.

4. When a part is confidently wrong

The Check part holds the draft against the six quality criteria from PRD §3 before a human sees it. A claim without a source tag is flagged. A source without an appraisal note is flagged. An unexplained technical term is flagged. If Check finds a failure, the draft goes back to the human reviewer with a check-report naming the failing line and the criterion it breaks — not back to Draft for an automatic rewrite, because the failure may be in the source, not in the sentence.

After Check, the Tester (Valentina) reads the draft. After her, the PM (Anisah) gives final sign-off. Both happen before Publish. This two-person gate is the step the PRD says is never automated.

Decision 6: a checking part reads first; then two named people read before publish. All three, in that order.

5. Trace to the PRD

Every promise in the PRD lands in a part, or we say why not:

PRD promise	Part	Status
Draw in the week's material, Dutch or English (PRD §2, step 1)	Intake	Covered
Retrieve further sources in both languages (PRD §2, step 2)	Retrieve + Translate	Covered
Appraise each source before use (PRD §2, step 3)	Appraise	Covered
Human review and edit (PRD §2, step 4)	Human reviewer (not a platform part)	By design — never automated
Publish the approved chapter (PRD §2, step 5)	Publish	Covered
Stop after each step, inspect, correct (PRD §2, final paragraph)	Orchestrator pauses after each stage; team inspects	Architectural — the pipeline is step-at-a-time
Six measurable quality criteria (PRD §3)	Check + Context (criteria live in context file)	Covered; becomes the evaluation pipeline in week 4
Non-coder must run it via CLI (PRD §4)	All parts triggered by natural-language CLI prompts	Covered

PRD promise	Part	Status
Dutch transcription retained for verification (PRD §4)	Translate keeps paired .nl + .en files	Covered
Manual fallback for every step (PRD §4 implicit)	Every part's contract is a file a person can produce by hand	Architectural, not a part
Never automate: interview, schedule decisions, sensitive data, approval (PRD §5)	These are out of scope for every part	Covered by exclusion + the human gate

6. Open questions

- **We do not yet know which ECD Thuiszorg Totaal runs.** If they use Nedap Ons with Caren already available, several external research questions change shape. The blueprint itself for the research platform does not change but the context file does, and with it what Retrieve searches for.
- **The context file is a stub.** It names the SME and the six criteria. What else goes in it including the weekly theme, which interview data the platform may reference is week 3's knowledge architecture decision.
- **Check is a checklist, not a pipeline.** The six criteria are checked by prompting the LLM to evaluate the draft against them. Adequate for week 2; becomes the automated evaluation pipeline in week 4, which will need calibration against human judgement.
- **Publishing mechanism unconfirmed.** We assume the handbook site accepts markdown via git push. If it is UI-only, Publish becomes a person pasting the file which is the manual fallback already described.
- **Thuiszorg Totaal is not yet formally confirmed** as our field-research partner. If the match changes, the context file changes; the pipeline does not.